

Numpy Lab – Part 1:

1. Create the array:

```
A = np.arange(1, 21).reshape(4, 5)
```

- Print: A, A.dtype, A.ndim, A.shape, A.size, A.itemsize
- Create B by reshaping A into shape (5, 4) and print B.

2. Given the following two arrays:

```
a = np.array([[0.,0.,0.],  
             [10.,10.,10.],  
             [20.,20.,20.],  
             [30.,30.,30.]])
```

```
b = np.array([0., 1., 2.])
```

- Compute $c = a + b$ and print it.
- Create a boolean array $mask = (c \geq 22)$ and print it.
- Count how many values in c are ≥ 22 .

3. Given the following array:

```
grades = np.array([[87, 96, 70],  
                  [100, 87, 90],  
                  [94, 77, 90],  
                  [100, 81, 82]])
```

Compute and print:

- Overall: sum, min, max, mean, std, var
- Mean per column (axis=0)
- Mean per row (axis=1)
- Max per row (axis=1)

4. Create a NumPy array of 2 X 3 dimensions. Initialize your array with the following numbers:

```
[2, 4, 6], [1, 3, 5], [7, 9, 11]
```

- Find the sum of each row
- Find the sum of each column
- Find the overall mean of the matrix

5. Given the following two arrays:

```
A = np.array([[10, 20, 30],  
             [40, 50, 60]])
```

B = np.array([1, 2, 3])

- **Add B to A**
- **Multiply A by 2**
- **Subtract 5 from all elements of A**

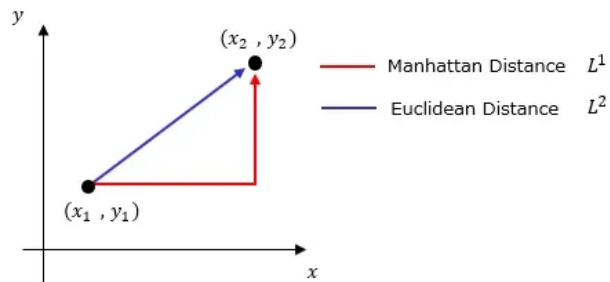
6. Given the following array:

```
grades = np.array([[70, 85, 90],  
                  [88, 76, 95],  
                  [60, 80, 72]])
```

- **Find the index of the largest value in the whole array**
- **Find the index of the highest grade in each row**
- **Find the index of the lowest grade in each column**

7. Manhattan and Euclidean Distance are computed as follow:

$$L^1 = |x_2 - x_1| + |y_2 - y_1|$$
$$L^2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Write two functions to compute the two types of distance